Mathematics for Cryptography II: Security Analysis and Polynomials

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Blockstream Research

research.blockstream.com

github.com/BlockstreamResearch/zk-book



Plan

1 Quick Recap

2 Security Analysis

3 Polynomials

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- How to read. . . This. . .

Definition 4 (Hiding Commitment)**.** *A commitment scheme is said to be hiding if for all PPT adversaries $\mathcal{A}$ there exists a negligible function $\mu(\lambda)$ such that.*

$$\left| \mathrm{P}\left[b = b' \;\middle|\; \begin{array}{l} \mathrm{pp} \leftarrow \mathrm{Setup}(1^\lambda); \\ (x_0, x_1) \in \mathsf{M}_{\mathrm{pp}}^2 \leftarrow \mathcal{A}(\mathrm{pp}), b \xleftarrow{\$} \{0,1\}, r \xleftarrow{\$} \mathsf{R}_{\mathrm{pp}}, \\ \mathbf{com} = \mathrm{Com}(x_b; r), b' \leftarrow \mathcal{A}(\mathrm{pp}, \mathbf{com}) \end{array} \right] - \frac{1}{2} \right| \leqslant \mu(\lambda)$$

where the probability is over $b, r,$ Setup *and $\mathcal{A}$. If $\mu(\lambda) = 0$ then we say the scheme is* perfectly hiding.

Definition 5 (Binding Commitment)**.** *A commitment scheme is said to be binding if for all PPT adversaries $\mathcal{A}$ there exists a negligible function μ such that.*

$$\mathrm{P}\left[\mathrm{Com}(x_0; r_0) = \mathrm{Com}(x_1; r_1) \wedge x_0 \neq x_1 \;\middle|\; \begin{array}{l} \mathrm{pp} \leftarrow \mathrm{Setup}(1^\lambda), \\ x_0, x_1, r_0, r_1 \leftarrow \mathcal{A}(\mathrm{pp}) \end{array} \right] \leqslant \mu(\lambda)$$

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Figure: This is not that hard as it seems. Figure from "*Bulletproofs: Short Proofs for Confidential Transactions and More*"

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3. Probability of event E is denoted by $\Pr[E]$, and $x \leftarrow\!\$\ S$ means that x is sampled uniformly at random from S.
4. $\#S$ is the number of elements in the finite set S.

Security Analysis

Introducing the Cipher: a bit of Formalism

We will consider an example of a cipher to demonstrate the notion of security.

Let us introduce three sets:

- $\mathcal{K}$ — a set of all possible keys.
- $\mathcal{M}$ — a set of all possible messages. For example, $\mathcal{M} = \{0,1\}^n$ — all binary strings of length n.
- $\mathcal{C}$ — a set of all possible ciphertexts.

The cipher is defined over a tuple $(\mathcal{K}, \mathcal{M}, \mathcal{C})$.

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Tiny Note

Cryptography is a very formalized field, but everything considered is well-known to you.

Introducing the Cipher

Definition

Cipher $\mathcal{E} = (E, D)$ over the space $(\mathcal{K}, \mathcal{M}, \mathcal{C})$ consists of two efficiently computable methods:

- $E : \mathcal{K} \times \mathcal{M} \to \mathcal{C}$ — encryption method, that based on the provided message $m \in \mathcal{M}$ and key $k \in \mathcal{K}$ outputs the cipher $c = E(k, m) \in \mathcal{C}$.
- $D : \mathcal{K} \times \mathcal{C} \to \mathcal{M}$ — decryption method, that based on the provided cipher $c \in \mathcal{C}$ and key $k \in \mathcal{K}$ outputs the message $m = D(k, c) \in \mathcal{M}$.

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Question

What else cipher must have to be practical?

Defining Security

Typically, the security is defined as a **game** between the adversary ($\mathcal{A}$) and the challenger ($\mathcal{C}h$).

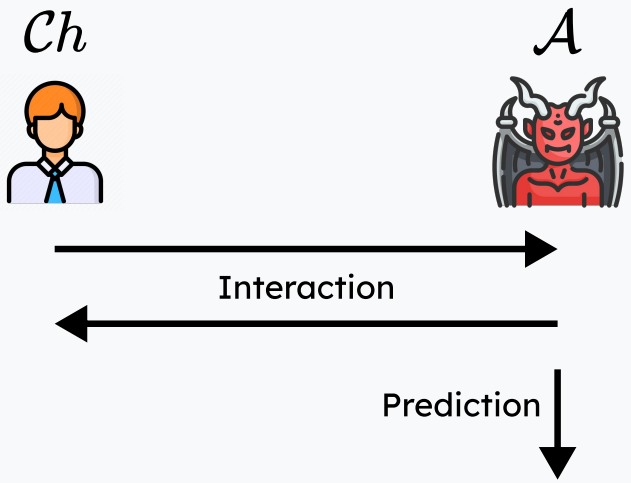


Figure: Challenger $\mathcal{C}h$ follows a straightforward protocol, while the adversary $\mathcal{A}$ might take any (efficient) strategy to win the game.

Efficient

Efficient means that an algorithm requires polynomial time and memory with respect to the security parameter (defined in a moment).

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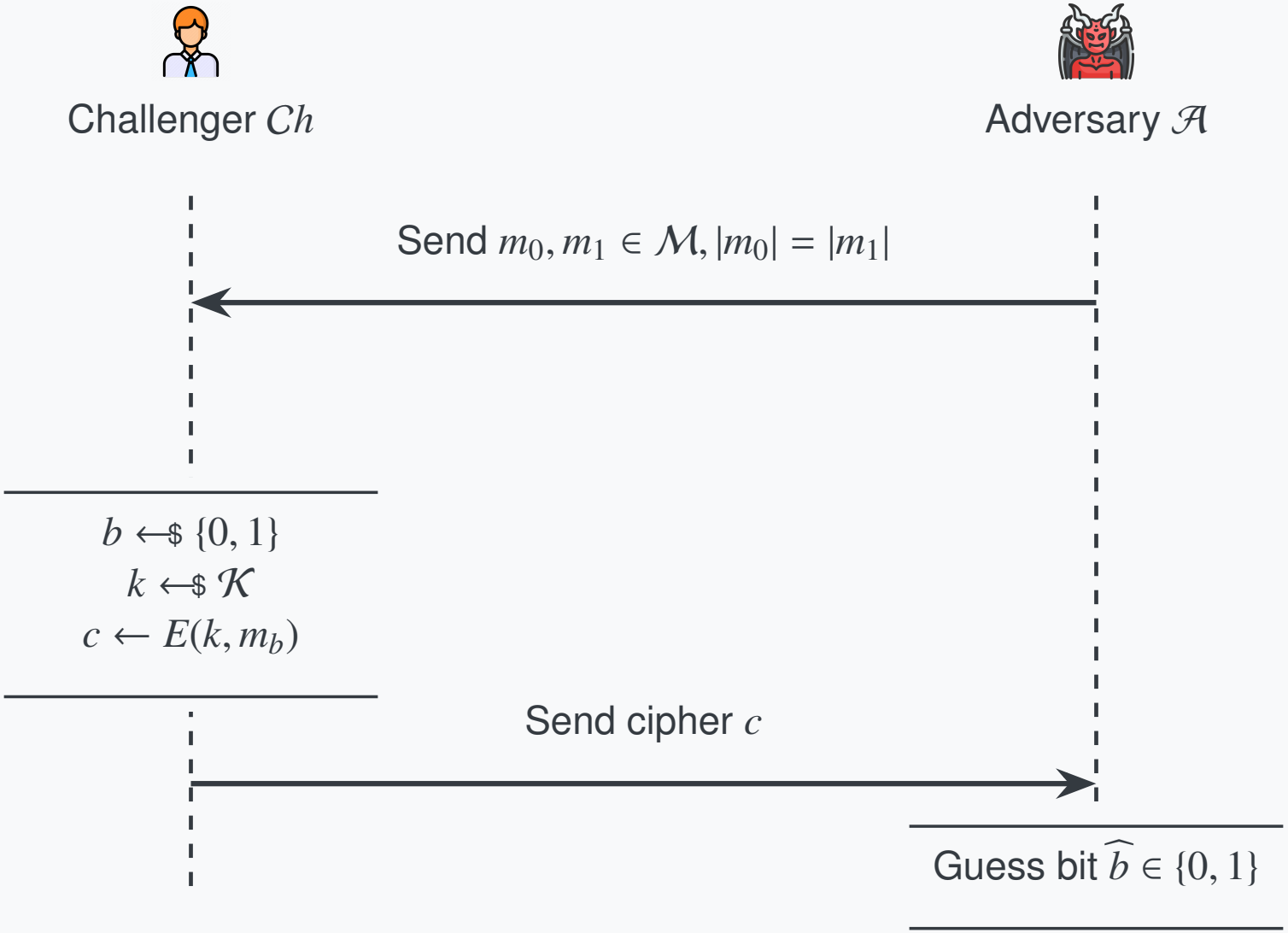
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4: $\mathcal{A}$ outputs the guess $\widehat{b}$.

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$$\mathsf{Adv}^{\mathsf{SS}}_{\mathcal{A},\mathcal{E}} := \left| \Pr[\widehat{b} = b] - \frac{1}{2} \right| .$$

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Definition

The cipher $\mathcal{E}$ is called **semantically secure** if for any efficient adversary $\mathcal{A}$ the advantage $\mathrm{Adv}^{\mathrm{SS}}_{\mathcal{A},\mathcal{E}}$ is **negligible**.

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Note

Advantage is just a measure of how much information is leaked to the adversary. The smaller the advantage, the better the cryptographic system. Formally, we want advantage to be **negligible**.

But what does it mean to be **negligible**?

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Security parameter, denoted by $\lambda \in \mathbb{N}$, is the "logarithmic size" of the scheme. We then treat the advantage as a function $\mathsf{Adv}^{\mathsf{SS}}_{\mathcal{A},\mathcal{E}}(\lambda) : \mathbb{N} \to \mathbb{R}_{\geq 0}$.

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Example

λ might correspond to the logarithm of the order of the underlying finite field or, for a cipher, we might set $\lambda :=$ "ciphertext size in bits".

Negligible Functions

Now, the probability of adversary winning the game depends on λ and we want this dependence to decrease rapidly.

Definition

A function $f : \mathbb{N} \to \mathbb{R}$ is called **negligible** if for all $\gamma \in \mathbb{R}_{>0}$ there exists $n_\gamma \in \mathbb{N}$ such that for any $n \geq n_\gamma$ we have $|f(n)| < 1/n^\gamma$.

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Less formally: f decays faster than any polynomial. The alternative definition, which is probably easier to interpret, is the following.

Theorem

A function $f : \mathbb{N} \to \mathbb{R}$ is **negligible** if and only if for any $\gamma \in \mathbb{R}_{>0}$, we have

$$\lim_{\lambda \to \infty} f(\lambda)\lambda^\gamma = 0.$$

Negligible Functions Examples

Example

The function $f(\lambda) = 2^{-\lambda}$ is negligible since for any $\gamma \in \mathbb{R}_{>0}$ we have $\lim_{\lambda \to \infty} 2^{-\lambda} \lambda^{\gamma} = 0$.
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Question

Is the function $u(\lambda) = \lambda^{-10000}$ negligible?

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How to read it

$\Pr[A \mid B]$ is "the probability of A, given that B occurred". On the right, we list the steps of the game; on the left, the event we are interested in.

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Definition (Security against message recovery)

Define $\mathsf{Adv}^{\mathsf{MR}}_{\mathcal{A},\mathcal{E}}(\lambda) := \left|\Pr[\widehat{m} = m] - \frac{1}{\#\mathcal{M}}\right|$. A cipher $\mathcal{E}$ is **secure against message recovery** if for all efficient adversaries $\mathcal{A}$, $\mathsf{Adv}^{\mathsf{MR}}_{\mathcal{A},\mathcal{E}}(\lambda) = \mathsf{negl}(\lambda)$.

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Theorem

If $\mathcal{E}$ is semantically secure, then $\mathcal{E}$ is secure against message recovery.

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$$(\mathrm{A} \Rightarrow \mathrm{B}) \iff (\neg\mathrm{B} \Rightarrow \neg\mathrm{A}).$$

So: given an efficient adversary $\mathcal{A}$ breaking message recovery, we build an adversary $\mathcal{B}$ breaking semantic security. $\mathcal{B}$ uses $\mathcal{A}$ as a black box (*oracle*), written $\mathcal{B}^{\mathcal{A}}$.

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One can show that $\mathrm{Adv}^{\mathrm{SS}}_{\mathcal{B},\mathcal{E}} = \mathrm{Adv}^{\mathrm{MR}}_{\mathcal{A},\mathcal{E}}$, so $\mathcal{B}$ breaks the semantic security with a non-negligible probability.

Security Reduction: the Recipe

Proposition (Security Reduction)

Suppose we have a cryptosystem $\mathcal{E}$ and we believe the problem $\mathcal{X}$ is computationally hard. To prove that $\mathcal{E}$ is secure based on $\mathcal{X}$:

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2: Construct an efficient algorithm $\mathcal{B}$ (the **reduction**) that solves $\mathcal{X}$ using oracle access to $\mathcal{A}$: on input x of the problem $\mathcal{X}$, $\mathcal{B}$ generates the input to the security game of $\mathcal{E}$, runs $\mathcal{A}$ and uses its output to solve $\mathcal{X}$.

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4: Hence assumption $\mathcal{X}$ implies security of $\mathcal{E}$.

Discrete Logarithm Assumption (DL)

Definition

Assume that $\mathbb{G}$ is a cyclic group of prime order $r = \Theta(2^\lambda)$ generated by $g \in \mathbb{G}$. Define the following game:

1: Both Ch and $\mathcal{A}$ take a description $\mathbb{G}$ as an input: order r and generator $g \in \mathbb{G}$.

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3: The adversary $\mathcal{A}$ outputs $\widehat{\alpha} \in \mathbb{Z}_r$.

We define $\mathcal{A}$'s **advantage in solving the discrete logarithm problem in** $\mathbb{G}$, denoted as $\mathsf{Adv}^{\mathsf{DL}}_{\mathcal{A},\mathbb{G}}$, as the probability that $\widehat{\alpha} = \alpha$.

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Definition

The **Discrete Logarithm Assumption** holds in the group $\mathbb{G}$ if for any efficient adversary $\mathcal{A}$ the advantage $\mathrm{Adv}^{\mathrm{DL}}_{\mathcal{A},\mathbb{G}}(\lambda)$ is negligible in λ.

Computational Diffie-Hellman Assumption (CDH)

Definition

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2: The adversary $\mathcal{A}$ outputs $\widehat{w} \in \mathbb{G}$.

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Computational Diffie-Hellman Assumption (CDH)

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Definition

The **Computational Diffie-Hellman Assumption** holds in the group $\mathbb{G}$ if for any efficient adversary $\mathcal{A}$, $\mathsf{Adv}^{\mathsf{CDH}}_{\mathcal{A},\mathbb{G}}(\lambda) = \mathsf{negl}(\lambda)$.

Decisional Diffie-Hellman Assumption (DDH)

Definition

Let $\mathbb{G}$ be a cyclic group of prime order $r = \Theta(2^\lambda)$ generated by $g \in \mathbb{G}$. Define the following game:

1: Ch computes $\alpha, \beta, \gamma \leftarrow\$ \mathbb{Z}_r$, $u \leftarrow g^\alpha, v \leftarrow g^\beta, w_0 \leftarrow g^{\alpha\beta}, w_1 \leftarrow g^\gamma$. Then, Ch flips a coin $b \leftarrow\$ \{0,1\}$ and sends (u, v, w_b) to $\mathcal{A}$.

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$$\mathsf{Adv}^{\mathsf{DDH}}_{\mathcal{A},\mathbb{G}}(\lambda) := \left| \Pr[b = \widehat{b}] - \frac{1}{2} \right|.$$

Decisional Diffie-Hellman Assumption (DDH)

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1: $\mathcal{C}h$ computes $\alpha, \beta, \gamma \leftarrow\$ \mathbb{Z}_r$, $u \leftarrow g^\alpha, v \leftarrow g^\beta, w_0 \leftarrow g^{\alpha\beta}, w_1 \leftarrow g^\gamma$. Then, $\mathcal{C}h$ flips a coin $b \leftarrow\$ \{0,1\}$ and sends (u, v, w_b) to $\mathcal{A}$.

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DDH is considerably stronger (and more fragile) than CDH. Let us show when DDH does not hold!

Even-ordered Cyclic Group

Theorem

Suppose that $\mathbb{G}$ is a cyclic group of an even order. Then, the Decisional Diffie-Hellman Assumption does not hold in $\mathbb{G}$. In fact, there is an efficient adversary $\mathcal{A}$ with an advantage $1/4$.

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Idea of proof. Let $\#\mathbb{G} = 2n$. We first prove the following statement:

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Suppose $u = g^{\alpha}$. Then, α is even if and only if $u^n = 1$.

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Therefore, if $u = g^{\alpha}, v = g^{\beta}, w = g^{\gamma}$, the adversary knows the parity of α, β, γ. He then checks if parity of $\alpha\beta$ equals parity of γ.

Polynomials

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A **polynomial** $p(T)$ is a function of the form

$$p(T) = c_0 + c_1 T + c_2 T^2 + \cdots + c_n T^n = \sum_{k=0}^{n} c_k T^k,$$

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Definition

A set of polynomials depending on T with coefficients in a field $\mathbb{F}$ is denoted as $\mathbb{F}[T]$, that is

$$\mathbb{F}[T] = \left\{ p(T) = \sum_{k=0}^{n} c_k T^k : c_k \in \mathbb{F},\ k = 0, \ldots, n \right\}.$$

Examples of Polynomials

Example

Consider the finite field $\mathbb{F}_3$. Then, some examples of polynomials from $\mathbb{F}_3[T]$ are listed below:

1. $p(T) = 1 + T + 2T^2$.
2. $q(T) = 1 + T^2 + T^3$.
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If we were to evaluate these polynomials at $1 \in \mathbb{F}_3$, we would get:

1. $p(1) = 1 + 1 + 2 \cdot 1 \bmod 3 = 1$.
2. $q(1) = 1 + 1 + 1 \bmod 3 = 0$.
3. $r(1) = 2 \cdot 1 = 2$.

More about polynomials

Definition

The **degree** of a polynomial $p(T) = c_0 + c_1 T + c_2 T^2 + \dots$ is the largest $k \in \mathbb{Z}_{\geq 0}$ such that $c_k \neq 0$. We denote the degree of a polynomial as $\deg p$. We also denote $\mathbb{F}[T]^{\leq m} := \{p \in \mathbb{F}[T] : \deg p \leq m\}$.

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Theorem

For any polynomials $p, q \in \mathbb{F}[T]$ and $n := \deg p$, $m := \deg q$, the following two statements are true:

1. $\deg(pq) = n + m$.
2. $\deg(p + q) = \max\{n, m\}$ if $n \neq m$, and $\deg(p + q) \leq m$ if $m = n$.

Roots of Polynomials

Definition

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Theorem

Let $p \in \mathbb{F}[T]$, $\deg p \geq 1$. Then, $\alpha \in \mathbb{F}$ is a root of p if and only if there exists a polynomial q (with $\deg q = \deg p - 1$) such that

$$p(T) = (T - \alpha)q(T).$$

Polynomial Division

Theorem

Given $f, g \in \mathbb{F}[T]$ with $g \neq 0$, there are unique polynomials $q, r \in \mathbb{F}[T]$ such that

$$f = q \cdot g + r,\ 0 \leq \deg r < \deg g.$$

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Example

Consider $f(T) = T^3 + 2$ and $g(T) = T + 1$ over $\mathbb{R}$. Then, we can write $f(T) = (T^2 - T + 1)g(T) + 1$, so the remainder of the division is $r \equiv 1$. Typically, we denote this as:

$$f \operatorname{div} g = T^2 - T + 1, \quad f \bmod g = 1.$$

The notation is pretty similar to one used in integer division.

Polynomial Divisibility

Definition

A polynomial $f \in \mathbb{F}[T]$ is called **divisible** by $g \in \mathbb{F}[T]$ (or, g **divides** f, written as $g \mid f$) if there exists $h \in \mathbb{F}[T]$ such that $f = gh$.

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Definition

A polynomial $f \in \mathbb{F}[T]$ is said to be **irreducible** in $\mathbb{F}$ if there are no polynomials $g, h \in \mathbb{F}[T]$ both of positive degree such that $f = gh$.

Polynomial Divisibility

Example

A polynomial $f(T) = T^2 + 16$ is irreducible in $\mathbb{R}$. Also $f(T) = T^2 - 2$ is irreducible over $\mathbb{Q}$, yet it is reducible over $\mathbb{R}$:
$f(T) = (T - \sqrt{2})(T + \sqrt{2})$.

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$f(T) = (T - \sqrt{2})(T + \sqrt{2})$.

Example

There are no polynomials over complex numbers $\mathbb{C}$ with degree more than 1 that are irreducible. This follows from the *fundamental theorem of algebra*. For example, $T^2 + 16 = (T - 4i)(T + 4i)$.

Interpolation

Question

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Theorem

Given $n+1$ distinct points $(x_0, y_0), \ldots, (x_n, y_n)$, there exists a unique polynomial p of degree at most n such that $p(x_i) = y_i$ for all $i = 0, \ldots, n$.

Illustration with two points

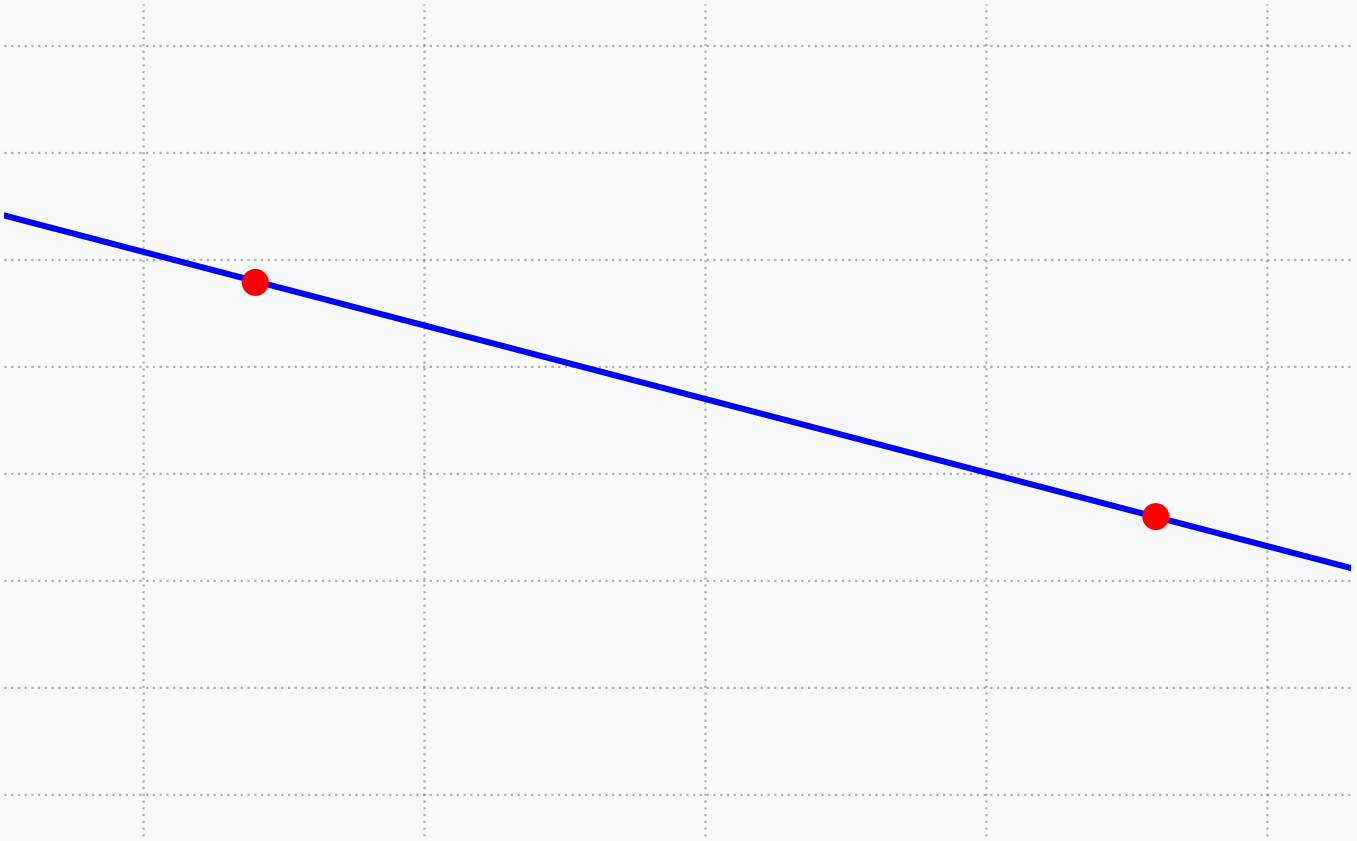


Figure: 2 points on the plane uniquely define the polynomial of degree 1 (linear function).

Illustration with five points

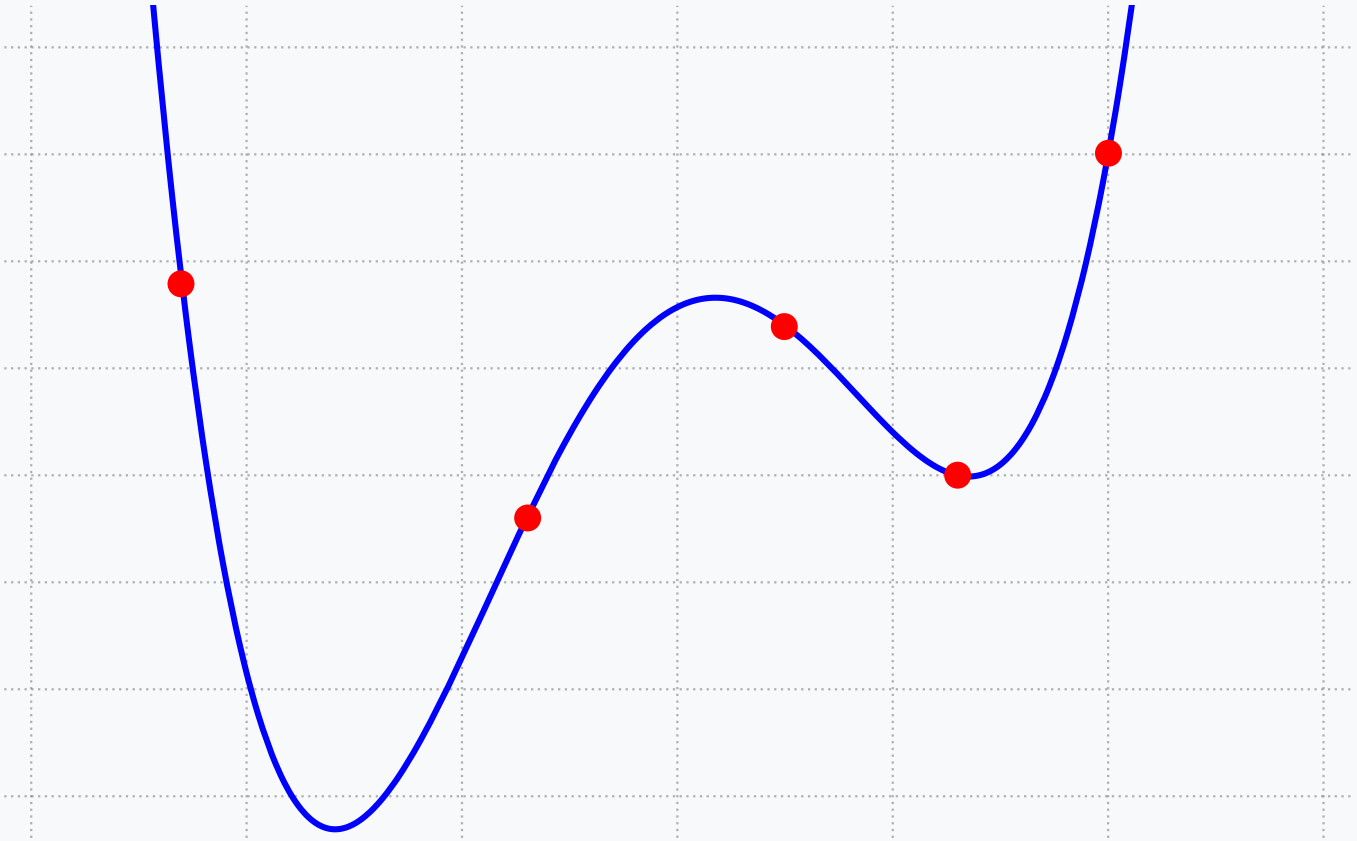


Figure: 5 points on the plane uniquely define the polynomial of degree 4.

Lagrange Interpolation

One of the ways to interpolate the polynomial is to use the Lagrange interpolation.

Theorem

Given $n+1$ distinct points $(x_0, y_0), \ldots, (x_n, y_n)$, the unique polynomial L of degree at most n that passes through these points is given by

$$L(T) = \sum_{i=0}^{n} y_i \ell_i(T), \quad \ell_i(T) = \prod_{j=0, j\neq i}^{n} \frac{T - x_j}{x_i - x_j}.$$

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Example

For points $(0, 1)$ and $(1, 2)$, we get

$$L(T) = 1 \cdot \frac{T-1}{0-1} + 2 \cdot \frac{T-0}{1-0} = T + 1.$$

Application: Shamir Secret Sharing

Motivation

How to share a secret α among n people in such a way that any t of them can reconstruct the secret, but any $t - 1$ cannot?

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Definition

Secret Sharing scheme is a pair of efficient algorithms $(\mathrm{Gen}, \mathrm{Comb})$ which work as follows:

- $\mathrm{Gen}(\alpha, t, n)$: probabilistic sharing algorithm that yields n shards $(\alpha_1, \ldots, \alpha_n)$ for which t shards are needed to reconstruct the secret α.
- $\mathrm{Comb}(\mathcal{I}, \{\alpha_i\}_{i \in \mathcal{I}})$: deterministic reconstruction algorithm that reconstructs the secret α from the shards $\mathcal{I} \subset \{1, \ldots, n\}$ of size t.

Shamir's Protocol

Correctness

For every $\alpha \in F$, for every possible output $(\alpha_1, \ldots, \alpha_n) \leftarrow \text{Gen}(\alpha, t, n)$, and any t-size subset $\mathcal{I}$ of $\{1, \ldots, n\}$ we require

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Shamir's protocol works over $F = \mathbb{F}_q$ as follows:

- $\mathsf{Gen}(\alpha, t, n)$: choose random $k_1, \ldots, k_{t-1} \leftarrow\!\!\$\ \mathbb{F}_q$ and define the polynomial

$$\omega(T) := \alpha + k_1 T + k_2 T^2 + \cdots + k_{t-1} T^{t-1} \in \mathbb{F}_q[T]^{<t},$$

and then compute $\alpha_i \leftarrow \omega(i) \in \mathbb{F}_q$ for each $i \in \{1, \ldots, n\}$.

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- $\mathsf{Comb}(\mathcal{I}, \{\alpha_i\}_{i \in \mathcal{I}})$: interpolate the polynomial ω using the Lagrange interpolation and output $\omega(0) = \alpha$.

Shamir's Protocol: Intuition

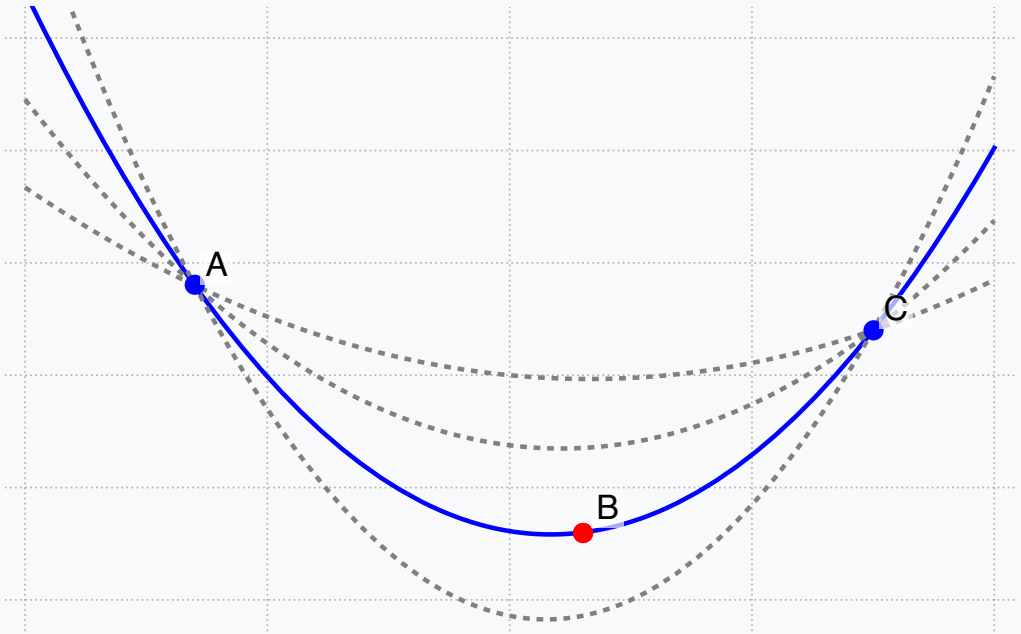


Figure: There are infinitely many quadratic polynomials passing through two blue points (gray dashed lines). However, knowing the red point allows us to uniquely determine the polynomial and thus get its value at 0.

Reed-Solomon Codes

Reed-Solomon codes

Reed-Solomon codes are an error-correcting algorithm based on polynomials. They allow restoring lost or corrupted data, implementing threshold secret sharing, and they are used in some ZK protocols (we will come back to them when building STARKs).

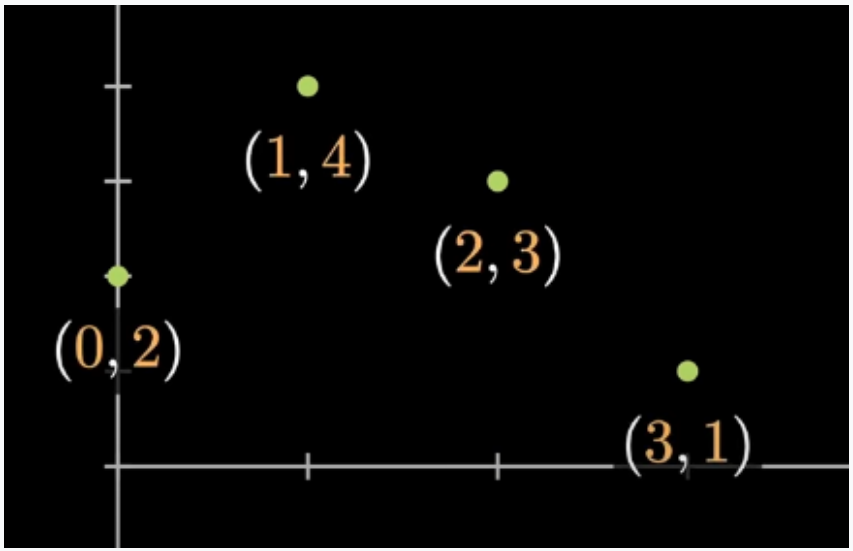


Figure: Polynomial of degree n is uniquely defined by $n + 1$ points. Defining more points on the same polynomial adds a redundancy, which can be used to restore the polynomial even if some points are missing.

Schwartz-Zippel Lemma

Lemma (Single Variable Schwartz-Zippel Lemma)

Let $\mathbb{F}$ be a finite field and let $f \in \mathbb{F}[T]$ be a non-zero polynomial. Then,

$$\Pr_{\alpha \leftarrow \$ \mathbb{F}}[f(\alpha) = 0] \leq \frac{\deg f}{\#\mathbb{F}}.$$

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Lemma (Multivariable Schwartz-Zippel)

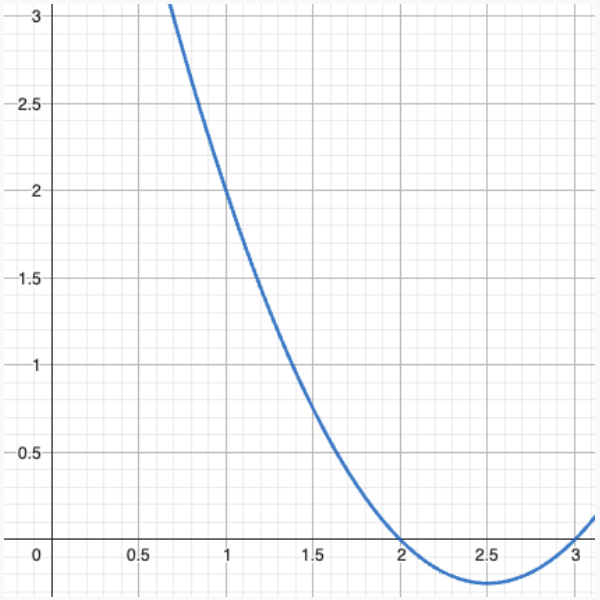
Let $S \subseteq \mathbb{F}^n$ be finite and f be a non-zero polynomial in n variables. Then,

$$\Pr_{\alpha \leftarrow \$ S}[f(\alpha) = 0] \leq \frac{\deg f}{\#S}.$$

Schwartz-Zippel Lemma: Example

Example

Suppose we are working over the Mersenne prime field of order $p = 2^{31} - 1$ with a polynomial $f(T, W) = T^{50} + 3TW + W^5$. If we pick (α, β) at random from $\mathbb{F}_p^2$, the probability of $f(\alpha, \beta) = 0$ is at most $50/(2^{31} - 1)$, providing approximately 25 bits of security.



Polynomial Equality Test

Corollary (Polynomial Equality Test)

Suppose P and Q are two distinct polynomials with a maximum degree d over the finite field $\mathbb{F}$. Then,

$$\Pr_{r \leftarrow \$ \mathbb{F}}[P(r) = Q(r)] \leq \frac{d}{\#\mathbb{F}}.$$

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Proof

Let $Z(T) := P(T) - Q(T)$. Of course, $\deg Z$ is at most d. Also note that $Z(r) = 0$ iff $P(r) = Q(r)$. Applying Schwartz-Zippel lemma, the probability of $Z(r) = 0$ for $r \leftarrow\$ \mathbb{F}$ is bounded above by $d/\#\mathbb{F}$. □

Thank you for your attention



research.blockstream.com

github.com/BlockstreamResearch/zk-book



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